

Lesson 4-1 · Period 2

Inverse Variation Applications + DOK-3 Performance Task

Klimesara-adapted · Student-centered · No Blooket

Do Now — Is this inverse variation?

Algebra 2 · Lesson 4-1 · Period 2

DOK 1

5 min

bridge from P1

Practice #15 · Table check

x	1	2	3	4	5	6
y	60	30	20	15	12	10

Is this an inverse variation? Compute xy for each column. If yes, state k .

DOK 2

12 min

teacher models

Application Rules for Inverse Variation

1. Identify two varying quantities and their constant PRODUCT k .
2. Compute k from one given pair.
3. Write the equation $y = k/x$ (rename variables to fit).
4. Solve for the unknown.

Example (bouzouki string): length s varies inversely with frequency f .

- 26-inch string $\rightarrow 329.63$ cycles/sec $\rightarrow k = 8,570.38$
- 13-inch string $\rightarrow f = 8,570.38/13 = 659.26$ cycles/sec

DIRECT ($y/x = k$) **vs INVERSE** ($xy = k$) — both appear in today's DOK-3!

Explore — TPS + DOK-3 Performance Task Algebra 2 · Lesson 4-1 · Period 2

DOK 2-3

33 min

student work

5 items in order. Plan ~15 min for Practice #26 (Ramón).

Try It 3 · Ice cube melting Time to melt varies inversely with air temperature.

Practice #12 · HOT: inverse SQUARE $y = k/x^2$. If x is multiplied by 4, what happens to y ?

Practice #17 · Radio wavelength (graph) w varies inversely with frequency f . Find f when $w = 375$ m.

DOK 2

5 min

DOK-3 reveal + Wed bridge

Sentence frame · Ramón “Part A is _____ variation because distance and time have a constant **RATIO** of _____.

Part B is _____ variation because gas and time have a constant **PRODUCT** of _____.”

Bridge to Period 3 (Wednesday) Tomorrow we graph $y = 1/x$ and its translations. Watch for: asymptotes, domain, range, intercepts. Period 3 is **assessment-critical** for the Topic 4 8-Q assessment.

DOK 1-2

3 min

not graded

Complete on your exit ticket

1. Today I learned that _____.
2. The biggest trap in Ramón's task was _____.
3. I distinguish direct from inverse variation by _____.